Overview of Bayesian network

Subject: Bayesian network

$$P(h \mid D) = \frac{P(D \mid h)\,P(h)}{P(D)}$$

1.

Graphical model Formally, Bayesian networks are directed acyclic graphs (DAGs) whose nodes represent variables in the Bayesian sense: they may be observable quantities, latent variables, unknown parameters or hypotheses. Each edge represents a direct conditional dependency. Any pair of nodes that are not connected (i.e. no path connects one node to the other) represent variables that are conditionally independent of each other. Each node is associated with a probability function that takes, as input, a particular set of values for the node's parent variables, and gives (as output) the probability (or probability distribution, if applicable) of the variable represented by the node. For example, if m {\displaystyle m} parent nodes represent m {\displaystyle m} Boolean variables, then the probability function could be represented by a table of 2 m {\displaystyle 2^{m}} entries, one entry for each of the 2 m {\displaystyle 2^{m}} possible parent combinations. Similar ideas may be applied to undirected, and possibly cyclic, graphs such as Markov networks.

2.

p (θ , φ ■ x) ∝ p (x ■ θ) p (θ ■ φ) p (φ) . {\displaystyle p(\theta ,\varphi \mid x)\propto p(x\mid \theta)p(\theta \mid \varphi)p(\varphi).}

3.

are equivalent: that is they impose exactly the same conditional independence requirements. A causal network is a Bayesian network with the requirement that the relationships be causal. The additional semantics of causal networks specify that if a node X is actively caused to be in a given state x (an action written as do(X = x)), then the probability density function changes to that of the network obtained by cutting the links from the parents of X to X, and setting X to the caused value x. Using these semantics, the impact of external interventions from data obtained prior to intervention can be predicted.

4.

d-separation This definition can be made more general by defining the "d"-separation of two nodes, where d stands for directional. We first define the "d"-separation of a trail and then we will define the "d"-separation of two nodes in terms of that. Let P be a trail from node u to v. A trail is a loop-free, undirected (i.e. all edge directions are ignored) path between two nodes. Then P is said to be d-separated by a set of nodes Z if any of the following conditions holds:

5.

Pr (G = T , S = T , R = T) = Pr (G = T ■ S = T , R = T) Pr (S = T ■ R = T) Pr (R = T) = 0.99 × 0.01 × 0.2 = 0.00198. {\displaystyle {\begin{aligned}\Pr(G=T,S=T,R=T)&=\Pr(G=T\mid S=T,R=T)\Pr(S=T\mid R=T)\Pr(R=T)\\&=0.99\times 0.01\times 0.2\\&=0.00198.\end{aligned}}}

6.

OpenBUGS – Open-source development of WinBUGS. SPSS Modeler – Commercial software that includes an implementation for Bayesian networks. Stan (software) – Stan is an open-source package for obtaining Bayesian inference using the No-U-Turn sampler (NUTS), a variant of Hamiltonian Monte Carlo. WinBUGS – One of the first computational implementations of MCMC samplers. No longer maintained.

7.

Local Markov property X is a Bayesian network with respect to G if it satisfies the local Markov property: each variable is conditionally independent of its non-descendants given its parent variables:

8.

Factorization definition X is a Bayesian network with respect to G if its joint probability density function (with respect to a product measure) can be written as a product of the individual density functions, conditional on their parent variables:

9.

where de(v) is the set of descendants and V \ de(v) is the set of non-descendants of v. This can be expressed in terms similar to the first definition, as

10.

P (X v = x v ■ X j = x j for each X j that is a parent of X v) {\displaystyle {\begin{aligned}&\operatorname {P} (X_{v}=x_{v}\mid X_{i}=x_{i}{\text{ for each }}X_{i}{\text{ that is not a descendant of }}X_{v}\,)\\[6pt]={}&P(X_{v}=x_{v}\mid X_{j}=x_{j}{\text{ for each }}X_{j}{\text{ that is a parent of }}X_{v}\,)\end{aligned}}}

11.

Pr (R = T ■ G = T) = 0.00198 T T T + 0.1584 T F T 0.00198 T T T + 0.288 T T F + 0.1584 T F T + 0.0 T F F = 891 2491 ≈ 35.77 % . {\displaystyle \Pr(R=T\mid G=T)={\frac {0.00198_{TTT}+0.1584_{TFT}}{0.00198_{TTT}+0.288_{TTF}+0.1584_{TFT}+0.0_{TFF}}}={\frac {891}{2491}}\approx 35.77\%.}

12.

P ■ (X 1 = x 1 , … , X n = x n) = ∏ v = 1 n P ■ (X v = x v ■ X v + 1 = x v + 1 , … , X n = x n) {\displaystyle \operatorname {P} (X_{1}=x_{1},\ldots ,X_{n}=x_{n})=\prod _{v=1}^{n}\operatorname {P} \left(X_{v}=x_{v}\mid X_{v+1}=x_{v+1},\ldots ,X_{n}=x_{n}\right)}

13.

A Bayesian network (also known as a Bayes network, Bayes net, belief network, or decision network) is a probabilistic graphical model that represents a set of variables and their conditional dependencies via a directed acyclic graph (DAG). While it is one of several forms of causal notation, causal networks are special cases of Bayesian networks. Bayesian networks are ideal for taking an event that occurred and predicting the likelihood that any one of several possible known causes was the contributing factor. For example, a Bayesian network could represent the probabilistic relationships between diseases and symptoms. Given symptoms, the network can be used to compute the probabilities of the presence of various diseases. Efficient algorithms can perform inference and learning in Bayesian networks. Bayesian networks that model sequences of variables (e.g. speech signals or protein sequences) are called dynamic Bayesian networks. Generalizations of Bayesian networks that can represent and solve decision problems under uncertainty are called influence diagrams.

14.

Several equivalent definitions of a Bayesian network have been offered. For the following, let G = (V,E) be a directed acyclic graph (DAG) and let X = (Xv), v ∈ V be a set of random variables indexed by V.

15.

Inference complexity and approximation algorithms In 1990, while working at Stanford University on large bioinformatic applications, Cooper proved that exact inference in Bayesian networks is NP-hard. This result prompted research on approximation algorithms with the aim of developing a tractable approximation to probabilistic inference. In 1993, Paul Dagum and Michael Luby proved two surprising results on the complexity of approximation of probabilistic inference in Bayesian networks. First, they proved that no tractable deterministic algorithm can approximate probabilistic inference to within an absolute error ■ < 1/2. Second, they proved that no tractable randomized algorithm can approximate probabilistic inference to within an absolute error ■ < 1/2 with confidence probability greater than 1/2. At about the same time, Roth proved that exact inference in Bayesian networks is in fact #P-complete (and thus as hard as counting the number of satisfying assignments of a conjunctive normal form formula (CNF)) and that approximate inference within a factor 2n1−■ for every ■ > 0, even for Bayesian networks with restricted architecture, is NP-hard. In practical terms, these complexity results suggested that while Bayesian networks were rich representations for AI and machine learning applications, their use in large real-world applications would need to be tempered by either topological structural constraints, such as naïve Bayes networks, or by restrictions on the conditional probabilities. The bounded variance algorithm developed by Dagum and Luby was the first provable fast approximation algorithm to efficiently approximate probabilistic inference in Bayesian networks with guarantees on the error approximation. This powerful algorithm required the minor restriction on the conditional probabilities of the Bayesian network to be bounded away from zero and one by 1 / p (n) {\displaystyle 1/p(n)} where p (n) {\displaystyle p(n)} was any polynomial of the number of nodes in the network, n {\displaystyle n} .